

JOE ALBERT NGIGI

Mechatronics & Automation Engineer | Industrial Controls · Robotics · IoT

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PROFILE

Mechatronics engineer with hands-on experience across power plant instrumentation & control, industrial automation, and collaborative robotics. WorldSkills gold medalist (Kenya, 2023) and international finalist (Lyon, 2024) in Mechatronics. Skilled in PLC/SCADA programming, high-voltage systems maintenance, ROS-based robotics, and embedded/IoT product development. Known for translating complex technical problems into safe, reliable, and well-documented solutions.

CORE SKILLS

- **Automation & Controls:** PLC programming (Siemens), SCADA/HMI development, SIPROTEC protection relays, LOTO safety procedures
- **Robotics:** ROS (Gazebo, RViz), xArm 7 collaborative robotics, motion planning, AR-assisted task visualization
- **Embedded & IoT:** Arduino, ESP32, STM32 (STM32CubeIDE), Raspberry Pi, PCB design, MQTT, Node-RED, Firebase
- **Electrical & Instrumentation:** MV/HV substation inspection, multimeters, oscilloscopes, frequency & capacitance meters, DC sources
- **Software:** Python, Flutter, Git/GitHub, dashboard & cloud integration
- **Languages:** English (C2), Swahili (native), French (A1)

WORK EXPERIENCE

Assistant Shift Technician (Instrumentation & Control)

Oct 2025 – Present

Thika Power | Thika, Kenya

- Monitor, calibrate, and maintain critical plant instrumentation (pressure, temperature, and level transmitters), ensuring accurate readings across manometers, thermometers, and digital indicators.
- Troubleshoot and repair plant electrical panels, replacing faulty contactors, transformers, fuses, and relays to restore functionality with minimal downtime.
- Inspect critical assets in MV/HV rooms and the HV substation using multimeters, frequency meters, capacitance meters, DC sources, and oscilloscopes for precise circuit diagnostics.
- Diagnose and repair automated control circuits, sensors, and pneumatic/hydraulic actuators, valves, and motors.
- Enforce plant safety compliance through rigorous Lockout/Tagout (LOTO) procedures and rapid emergency response to protect personnel and equipment.
- Analyze operational data trends to help the engineering team predict failures and optimize plant performance.
- Contributed to a 24,000-hour Preventive Maintenance Service (PMS) cycle on a 16 MW generator, including testing and calibration of SIPROTEC protection relays.
- Manage inventory and lifecycle tracking of electrical tools/materials and provide internal IT network support.

Robotics Research Assistant

Sep 2025 – Jan 2026

Virtual Mechatronics Lab | Nyeri, Kenya

- Programmed and optimized an xArm 7 for high-precision human-robot collaborative (HRC) tasks in shared industrial workspaces.
- Refined motion planning and safety parameters to enable seamless human-robot interaction.
- Designed synchronized human-robot task workflows that improved operational efficiency and ergonomic safety.
- Collaborated on integrating Augmented Reality (AR) tools for real-time task visualization and operator awareness.
- Led testing and validation of collaborative robotics scenarios.

Embedded Systems Designer (Freelance)

Sep 2023 – Present

Fiverr | Remote

- Deliver IoT and embedded software projects for clients worldwide using ESP32, Arduino, and Raspberry Pi.

- Build cloud-connected dashboards and firmware with MQTT, Node-RED, and Firebase; maintain consistently high client ratings.

Flutter Mobile App Tutor

Mar 2024 – Aug 2024

Bynahry | Remote

- Designed and led an interactive Flutter course for kids, building a full curriculum from UI design through deployment.
- Increased student engagement by 50% through hands-on, project-based teaching.

Embedded Systems Instructor & Engineer

Jan 2023 – Apr 2023

ArtSam and Wira Technologies | Kenya

- Guided students in building STM32-based line-tracking robots and a 5-DOF robotic arm.
- Taught embedded systems fundamentals through hands-on STM32CubeIDE lessons.

SELECTED PROJECTS

- **Autonomous Mobile Robot (2023–2024):** Built a ROS2-based robot using Raspberry Pi 5, Lidar SLAM mapping, and an STM32-driven differential drive for autonomous waypoint navigation. [GitHub](#)
- **Festo Distributing & Sorting Station (2023):** Assembled and programmed a PLC1500/HMI/Scalance station to sort items by color and material for WorldSkills Lyon preparation. [GitHub](#)
- **xArm 7 Cowork (2025–Present):** Developed motion planning and safety logic for human-robot collaborative tasks. [GitHub](#)
- **PLC Elevator Digital Twin (2024):** Built a real-time digital twin of an elevator control system pairing custom hardware with simulation. [GitHub](#)
- **IoT Relay Control System (2024):** Created a globally accessible IoT control system using ESP32 endpoints, MQTT (HiveMQ), Node-RED dashboards, and Cloudflare tunneling.
- **STM32 Line-Tracking Robot (2024):** Developed a multifunctional line-tracking robot with an integrated robotic arm. [GitHub](#)
- **Rainfall Simulator (Thesis Project):** Built an Arduino Nano-based water flow control system with a motorized ball valve and a real-time Processing UI. [GitHub](#)

EDUCATION

BSc, Mechatronics Engineering — Second Class Upper

Sep 2020 – May 2025

Dedan Kimathi University of Technology | Nyeri, Kenya

Student Exchange Program — Generalist Informatique

Feb 2025 – Jul 2025

CESI Lyon | Lyon, France

- Selected for a 6-month project-based academic exchange covering AI, automation, and web development.
- Co-developed CESI Eats, a food delivery platform prototype, and modeled a 3D gluing automation system using AR-assisted layout optimization.

High School Diploma — Grade: A-

2016 – 2019

Nairobi School | Nairobi, Kenya

CERTIFICATIONS & AWARDS

- WorldSkills Lyon 2024 Champion — Mechatronics (International, France, 2024)
- Gold Medalist, WorldSkills Kenya — Mechatronics (National Competition, 2023)
- Siemens Mechatronics Systems Certification Program – Associate (Siemens Center, DeKUT, 2023)
- Siemens Certified Mechatronic Systems Assistant – Level 1 / SMSCP (Siemens Center, DeKUT, 2022)
- IoT Certification Course — HUBiquitous / IEEE DeKUT / Dedan Innovators Club (2024)
- Python Core - Sololearn
- Technical Award Winner -robotics Dojo 2024